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Models and algorithms for planning and scheduling problems

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Scheduling at Villa Vigoni

This issue contains a selection of the papers presented at the Workshop on “Models and Algorithms for Planning and Scheduling Problems” which was held in Loven di Menaggio, on the Lake of Como, 14–18 June 1993.

The workshop was put up in the Oberwolfach style, i.e., a limited number of participants (45) and lectures (35), surveys of experts in the field, ample time for discussion, the right mixture of senior researchers and young promising talents, and an inspiring scientific and social atmosphere.

This atmosphere was more than achieved by the splendid location in Villa Vigoni, a beautiful 19th century villa in neoclassical style on the hills of the Lake of Como. The Villa Vigoni is famous for its valuable interiors, its beautiful park and surroundings, and its excellent cuisine and hospitality.

The topics presented spanned the whole field of planning and scheduling. The major topics were:

- machine scheduling
- project analysis and scheduling
- vehicle scheduling and routing
- graph-theoretic models
- complexity of scheduling problems
- approximation algorithms
- on-line algorithms
- computational methods and software

A number of prominent people in the area were invited to give selected contributions: J. Blazewicz, G. Gallo, C.R. Glassey, D.S. Hochbaum, E.L. Lawler, J.K. Lenstra, M. Lucertini, F. Maffioli, M. Queyranne, D.B. Shmoys, P. Toth, G. Weiss, and D. de Werra.

Moreover, in order to have a high quality standard, the 24 contributed papers had to pass a strong selection process.

This issue includes 9 papers which cover several aspects of the research activities on scheduling problems: modelling techniques, $\mathcal{NP}$ -hardness results, polynomial algorithms, approximate algorithms and heuristics.

- The modelling issue is dealt with in the paper of Bampis, Guinand and Trystram who present and analyse scheduling problems that arise in the context of parallel programming. The relevance of communication delays in different parallel architectures implies the need for the extension of known models.

- Orman and Potts give an extensive and detailed complexity analysis of the problem to sequence jobs consisting of two operations on a single machine in order to minimize the makespan.

- The complexity of a different single machine problem with precedence constraints is studied by Steiner. The problem of minimizing the number of tardy jobs is shown to be $\mathcal{NP}$ -hard when tardy means that a job is not completed within a fixed time after its earliest possible completion time. Polynomial cases are obtained for special classes of precedence constraints.
- The problem of scheduling multiprocessor tasks, i.e. tasks which require a given set of processors, with the objective of minimizing the makespan is considered by Dell’Olmo, Speranza and Tuza. This problem, which is known to be $\mathcal{NP}$ -hard, is equivalent to the graph-theoretic interval coloring problem. It is shown that the problem remains $\mathcal{NP}$ -hard on some specific classes of perfect graphs, but that solutions with a constant performance ratio can be determined efficiently.
- Hochbaum and Landy present a polynomial algorithm that improves the complexity of the best-known algorithm for scheduling batches of two job types in the presence of setup times such as to minimize the weighted sum of the completion times.

The design and analysis of approximate and heuristic algorithms is the basic issue of the last four papers. While two papers present approximation results for specific scheduling problems, the two remaining papers present heuristic approaches to relatively large classes of scheduling problems.

- Dell’Olmo et al. present linear time algorithms for the preemptive scheduling of multiprocessor tasks on two, three and four processors with the objective to minimize the maximum lateness. They identify cases for which these algorithms are optimal and obtain general performance guarantees.
- Gallo and Piccinonno obtain a tight worst-case bound for a new heuristic algorithm for scheduling tasks with tree structured precedence constraints on two identical machines with the objective to minimize the makespan.
- Local search techniques are experimentally shown to be very effective in the paper by Brucker, Huřink and Werner. They consider a class of scheduling problems that includes, among others, the classical parallel machine scheduling problem.
- Finally, a new algorithmic approach is presented in the paper by Jurisch, Kubiak and Jozefowska for a class of scheduling problems which can be reduced to the problem of finding a minimum weighted clique in a complete weighted graph.

The “Villa Vigoni” Workshop has turned out as the start of a series of workshops in the same spirit. The second such workshop took place in Wernigerode (Germany) in 1995 and the third workshop is planned to be held in England in 1997.

The workshop would not have been possible without significant support from several organizations. We would like to express our gratitude to the Deutsche Forschungsgemeinschaft, the Forschungsinstitut für Anwendungsorientierte Wissensverarbeitung (FAW) in Ulm, the Consiglio Nazionale delle Ricerche, Associazione Italiana di Ricerca Operativa, and the German–Italian Center at Villa Vigoni.

We also want to express our appreciation to the participants, the invited lecturers, the authors and all the referees for their contributions and reviews.

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